

Data Deduplication *with* a Linux Based File System

Data deduplication is a specialised technique for compressing data and deleting copies of repeated data. It plays an important role in today's world of rapid and massive data generation, as it helps to save resources, energy and costs. This article explains how Lessfs, a Linux based file system, can be used for data deduplication.



The existence of copies of the same files in different locations creates various management problems. One of the main problems, which is also present when it comes to simple storage systems, is data duplication. Storage space availability in most systems is used up by copies of the same files. As an example, the WhatsApp messenger app saves a different copy of the same image when it is received from different chats or forwarded to different persons. This reduces the space available on a device. This is where data deduplication comes in.

Data deduplication is a data compression technique to eliminate redundant data and decrease the space used up on an enabled storage volume. A volume can refer to a disk device, a partition or a grouped set of disk devices—all represented as a single device. During the process, redundant data is deleted, and a single copy of the data is stored on the storage volume.

Necessity for and merits of data deduplication

The primary focus of data deduplication is to point out large sections of data (which can include entire files or large sections of files) that are identical, and store only one copy of this data. Other benefits are:

- Reduced cost of storage equipment
- Reduced cost of energy
- Lesser need for cooling

There are two types of data deduplication — post-process deduplication and inline deduplication.

Post-process deduplication: In this method, the deduplication process starts after the data is stored. After the files are stored, the program checks for duplicated data throughout the file system and makes sure that only one copy exists. This method becomes problematic when the available space is already low and does not allow multiple copies of